

Development of the Monash True-Triaxial Hopkinson Bar System and an Integrated Computational Workspace for Multiaxial Dynamic Rock Characterisation

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Abstract

Conventional split-Hopkinson pressure bar (SHPB) testing characterises rock under one-dimensional dynamic loading, yet rock in deep mining, tunnelling and rock blasting fails under *multiaxial* stress states at high strain rate. This paper presents the development of the Monash triaxial Hopkinson bar (Tri-HB) — a gas-gun-driven system of three independent pairs of orthogonal square bars with hydraulic confinement on all three axes, enabling true-triaxial ($\sigma_2 \neq \sigma_3$) static–dynamic coupled loading — and an integrated *computational workspace*, the Virtual Tri-HB, that unifies test design, three-wave data reduction, stress-path interpretation and damage modelling in a single six-step workflow. We report six advances embedded in the computational workspace: (i) a unified treatment of uniaxial, axisymmetric-confined and true-triaxial static–dynamic loading as special cases of a single gas-gun system, reduced through a consistent per-axis three-wave analysis; (ii) a true-triaxial, Lode-angle-dependent and rate-dependent failure envelope that captures the intermediate principal stress effect with the correct sign; (iii) an energy-consistent damage–dissipation partition in which the first law closes by construction with non-negative dissipation, $W_{\text{input}} = W_{\text{el}} + W_{\text{diss}}$ ($W_{\text{diss}} \geq 0$); (iv) an orthotropic damage tensor whose components are driven chiefly by the directional tensile strain (with a small shear/compaction term), reproducing the directional (axial-splitting) fracture of rock blasting with a modest, nonzero axial damage; (v) a cumulative-damage model for repeated (cyclic) dynamic loading that reproduces the sudden, few-impact ($N_f \lesssim 10$) dynamic-fatigue failure of a pre-stressed rock mass; and (vi) a spatial thermodynamic extension comprising a gradient-regularised damage field (internal length ℓ calibrated from CT crack spacing) and a phase-field formulation (Griffith energy G_c from the measured dissipation and CT fracture area, phase-field $\phi(\mathbf{x})$ validated against CT attenuation maps) that is thermodynamically consistent by construction and directly comparable to post-impact CT cross-sections. Because the three orthogonal bars measure direction-resolved stress, strain and wave speed, each component of the damage tensor can in principle be calibrated independently — a capability a uniaxial SHPB cannot provide. We close with a calibrate-then-blind-predict validation roadmap that positions the Tri-HB and its computational workspace for quantitative, multiaxial dynamic constitutive modelling of rock.

Keywords: triaxial Hopkinson bar; dynamic rock mechanics; true-triaxial loading; intermediate principal stress; anisotropic damage; rock blasting; computational workspace; gradient damage; phase-field fracture; CT calibration.

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1 Introduction

The dynamic mechanical response of rock governs the safety and efficiency of deep underground excavation, rockburst mitigation and rock blasting, where loading rates of 10^1 – 10^3 s $^{-1}$ coincide with confining stresses of tens to hundreds of megapascals (Zhang and Zhao, 2014; Wang and Zhang, 2026). Capturing this response in the laboratory has driven more than a century of development of the Hopkinson bar, yet the classical apparatus remains one-dimensional. This section traces that development, examines how confinement has conventionally been added, and motivates a true-triaxial alternative.

1.1 From the Hopkinson bar to the split-Hopkinson pressure bar

The technique originates with Bertram Hopkinson, who in 1914 used a long elastic bar terminated by a momentum trap to record the pressure–time history produced by explosive detonation and bullet impact (Hopkinson, 1914) — the original *Hopkinson pressure bar*. Davies (1948) replaced the mechanical momentum measurement with electrical (condenser) sensing of bar displacement and gave the first rigorous treatment of geometric wave dispersion. Kolsky (1949) then split the bar in two and placed a thin specimen between an incident and a transmission bar: the incident, reflected and transmitted pulses recovered from bar gauges yield the specimen stress, strain rate and strain, defining the split-Hopkinson pressure bar (SHPB), or Kolsky bar, still in use today. Subsequent maturation added resistance strain-gauge instrumentation, Pochhammer–Chree dispersion correction, and pulse shaping to enforce a constant strain rate and stress equilibrium in brittle specimens (Frew et al., 2002); the method was extended to tension, torsion and elevated temperature, and codified for rock through suggested methods and reviews (Zhou et al., 2012; Zhang and Zhao, 2014; Wang and Zhang, 2026). Throughout, however, the loading remained *one-dimensional*: a single axial pulse on an unconfined specimen.

1.2 Confined and triaxial-chamber dynamic testing

Because rock in situ is confined, dynamic testing under lateral pressure followed soon after. Two strategies became standard. In *passive* confinement a rigid metal sleeve or shrink-fit ring surrounds the specimen and supplies a reactive radial constraint that rises with the axial load but is not independently controlled (Christensen et al., 1972). In *active* confinement the SHPB specimen is enclosed in a pressure cell and a hydraulic or pneumatic medium imposes a prescribed confining pressure before the dynamic axial pulse arrives (Lindholm et al., 1974). These confined-SHPB methods established the pressure dependence and dynamic increase of rock strength and remain widely used; they correspond to the confinement-chamber mode (Mode 2) of the present workflow. They share, however, three intrinsic limitations. First, the confinement is *axisymmetric*, $\sigma_2 = \sigma_3$, so the *intermediate principal stress effect* — shown by quasi-static true-triaxial testing to be first-order for rock strength (Mogi, 1971) — cannot be accessed. Second, the confinement is quasi-static and pre-applied; the lateral stress can be neither pulsed nor timed relative to the axial event. Third, the specimen is enclosed, so the lateral stress and strain are inferred through the cell rather than measured directly, and the lateral dynamic (Poisson-driven) response is not resolved.

From a boundary-value standpoint, the confining medium acts as a low-impedance *traction* boundary rather than a displacement constraint. On the wetted specimen surface S with outward normal $\mathbf{n}$, an inviscid fluid transmits only a normal pressure and couples to the wall acceleration through

$$\sigma_{ij}n_j = -p(t)n_i, \quad \frac{\partial p}{\partial n} = -\rho_f \ddot{u}_n, \quad (1)$$

the first a Neumann (traction) condition that leaves the lateral displacement free, the second the fluid–structure coupling that fixes any radiated energy. The acoustic mismatch with the rock, $R_{\text{lat}} = (Z_f - Z_s)/(Z_f + Z_s)$ with $Z = \rho c$, is severe — $Z_f/Z_s \sim 10^{-1}$ for liquids and $\lesssim 10^{-3}$ for

gases — so the surface behaves almost as a pressure-release boundary and radiates little, while the chamber pressure stays effectively quasi-static, $p \approx p_c$, over the $\sim 100 \mu\text{s}$ event. The confinement is thus a clean axisymmetric pressure ($\sigma_2 = \sigma_3 = p_c$) that neither suppresses lateral expansion nor yields a measurable lateral signal.

These chamber effects can be mitigated, and the relations above prescribe how. The dynamic pressure rise, $\Delta p \approx K_{\text{eff}}(V_s/V_f) 2\nu\varepsilon_{\text{ax}}$, is held well below p_c by a large reservoir ($V_f \gg V_s$) or a soft medium — a gas, with $K_{\text{eff}} = \gamma p_c$, is about two orders of magnitude better than hydraulic oil. Radiation into the medium is suppressed by smooth, long-wavelength loading ($ka \ll 1$, the same condition that controls bar dispersion); wall reflections are excluded from the measurement window t_w by sizing the chamber radius $d > c_f t_w/2$; and parasitic load paths are removed with a thin, soft sealing jacket ($A_j E_j \ll A_s E_s$) and non-load-bearing seals. Each remedy, however, only *mitigates* a low-impedance, enclosed boundary, and none recovers the lateral signal — the gap the next section closes by replacing the chamber with matched-impedance bars.

1.3 The case for a true-triaxial Hopkinson bar

The Monash triaxial Hopkinson bar (Tri-HB) removes the confining chamber altogether and instead loads a cubic specimen through *three independent pairs of orthogonal Hopkinson bars*, with hydraulic cylinders applying independent quasi-static pre-stress on each axis and a gas gun superposing the dynamic pulse on the X axis. Each limitation above becomes a capability. (i) *True-triaxial* states, $\sigma_1 > \sigma_2 > \sigma_3$, are imposed by setting the three pre-stresses independently, so the intermediate-stress and Lode-angle dependence can be probed dynamically. (ii) *Static-dynamic coupling* reproduces the in-situ condition of a pre-stressed rock mass struck by a dynamic disturbance, as in blasting or rockburst. (iii) The lateral response is *measured directly*: each transverse axis carries its own output bars and gauges (six gauge sets in total), so σ_y , σ_z and the directional fracture are resolved rather than inferred.

These hardware capabilities become quantitative only if the interpretation keeps pace. Reducing a multiaxial dynamic test to a constitutive statement requires a consistent three-wave analysis on each axis, a failure surface that depends on all three stress invariants, an energy-consistent dissipation balance, and a damage description that is *directional* — because blasting produces oriented cracks rather than isotropic softening. These ingredients are rarely assembled in one auditable workflow.

1.4 Scope and outline

This paper presents the Tri-HB system together with an integrated computational workspace that couples the hardware data-reduction equations to such a constitutive framework. Section 2 describes the hardware and the loading configurations it realises; Section 3 the one-dimensional wave propagation and the per-axis three-wave reduction; and Section 4 the integrated computational workspace. Section 5 presents the four constitutive advances — the true-triaxial rate-dependent envelope, the energy-consistent dissipation, the orthotropic damage tensor, and the cyclic cumulative-damage model. Section 6 gives the validation roadmap, and Section 7 concludes. All figures are generated by the Virtual Tri-HB workspace from prescribed-pulse reference cases; the present contribution is the system and its modelling framework, with experimental calibration identified as the immediate next step.

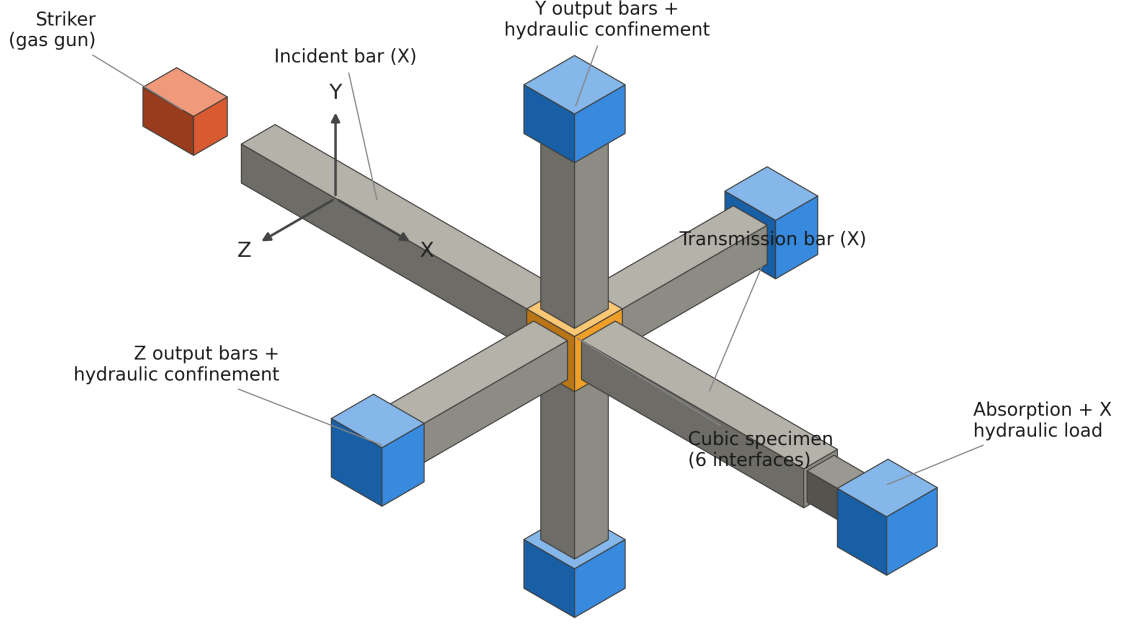


Figure 1: Isometric schematic of the Monash Tri-HB. The gas gun drives the X incident/transmission train; two output-bar pairs on Y and Z and a hydraulic load cylinder on X apply independent confinement, giving static–dynamic coupled true-triaxial loading on the central cubic specimen.

2 The Monash Tri-HB system

2.1 Hardware configuration

The Tri-HB (Fig. 1) couples a gas-gun dynamic loading train on the X axis with three independent pairs of orthogonal square bars and hydraulic confinement on all three axes. The dynamic train comprises a cylindrical striker ($\varnothing 40$ mm, length 0.5 m), an incident bar (2.5 m), a transmission bar (2 m), an absorption bar and a momentum trap, all of 42CrMo steel (Table 1). The lateral Y and Z axes each carry two 2 m square output bars. A cubic specimen is sandwiched at the centre, producing six rock–bar interfaces with no bar-to-bar contact; support wheels constrain each bar to its own axis. Three hydraulic cylinders (capacity up to 100 MPa) apply independent quasi-static pre-stress on X , Y and Z , while the gas gun (impact velocity up to 50 m/s) superposes the dynamic pulse on X . The result is genuine static–dynamic coupled, true-triaxial loading: the pre-stress sets a multiaxial initial state and the striker drives the high-rate event. Six sets of strain gauges (sampling at 1 MHz) on the incident, transmission and four output bars record the incident, reflected, transmitted and lateral-output strains, ε_{in} , ε_{re} , ε_{tr} , $\varepsilon_{y1,y2}$, $\varepsilon_{z1,z2}$.

2.2 Loading configurations

Because each axis carries an independent hydraulic pre-stress, the gas-gun Tri-HB realises the conventional dynamic-loading cases as special cases of a single apparatus (Table 2): the historical uniaxial SHPB when the lateral axes are unconfined (Mode 1), axisymmetric confined SHPB when the two lateral pre-stresses are equal (Mode 2), and genuine true-triaxial static–dynamic loading when they differ, $\sigma_2 \neq \sigma_3$ (Mode 3). Mode 3 is the configuration that distinguishes the system; Modes 1 and 2 are recovered from it simply by removing or equalising the lateral

Table 1: Representative Monash Tri-HB bar and loading parameters (42CrMo steel).

Quantity	Symbol	Value
Bar density	ρ_b	7850 kg/m ³
Bar Young’s modulus	E_b	210 GPa
Bar wave speed	C_0	5200 m/s
Bar yield strength	σ_y	930 MPa
Striker (diameter, length)	$\varnothing, L_s$	40 mm, 0.5 m
Square-bar cross-section	A_b	50 mm $\times$ 50 mm
Incident / transmission / output bar	L	2.5 / 2.0 / 2.0 m
Maximum impact velocity	v	50 m/s
Hydraulic confinement capacity	p_c	100 MPa
Gauge sampling rate	—	1 MHz

Table 2: Loading configurations of the Monash Tri-HB, set by the per-axis hydraulic pre-stress.

Mode	Configuration	Lateral pre-stress	True-triaxial
1	Uniaxial SHPB	none ($\sigma_2 = \sigma_3 = 0$)	no
2	Axisymmetric confined SHPB	equal ($\sigma_2 = \sigma_3 > 0$)	partial
3	Tri-HB static–dynamic	unequal ($\sigma_2 \neq \sigma_3$)	yes

confinement, which is what makes uniaxial, confined and true-triaxial results directly comparable on one apparatus.

The corresponding p – q stress paths (Fig. 2) are plotted in *total* specimen stress, $\sigma = \sigma_{\text{pre}} + \Delta\sigma_{\text{dyn}}$, not in dynamic increments. The uniaxial case carries no pre-stress and therefore starts at the origin, whereas the true-triaxial case — here pre-stressed to $(\sigma_1, \sigma_2, \sigma_3) = (30, 20, 15)$ MPa, i.e. a static point $(p_0, q_0) \approx (21.7, 13.2)$ MPa — begins at that point and moves outward as the dynamic X pulse is superposed. The bar-gauge reduction itself operates on the dynamic increments alone (§3.3); the total state is plotted because specimen failure is governed by the absolute stress $\sigma_{\text{pre}} + \Delta\sigma_{\text{dyn}}$, not by the increment, and this choice is what makes the pre-stress offset of the true-triaxial path physically meaningful.

3 Wave propagation and per-axis data reduction

Sign and notation conventions. We adopt the compression-positive convention standard in rock dynamics: normal stress and strain are positive in compression, and the particle velocity v is positive along $+x$, the incident-pulse propagation direction. The recorded bar-gauge strains are accordingly $\varepsilon_{\text{in}} > 0$ (compressive incident), $\varepsilon_{\text{re}} < 0$ (tensile reflected, for a specimen softer than the bars) and $\varepsilon_{\text{tr}} > 0$ (compressive transmitted). Principal stresses are ordered $\sigma_1 \geq \sigma_2 \geq \sigma_3$ and $\langle x \rangle = \max(x, 0)$ denotes the Macaulay bracket.

3.1 One-dimensional wave propagation and dispersion

Each pressure bar is a long, slender rod in which, for wavelengths much greater than the bar diameter, the axial displacement $u(x, t)$ obeys the one-dimensional elastic wave equation

$$\frac{\partial^2 u}{\partial t^2} = C_0^2 \frac{\partial^2 u}{\partial x^2}, \quad C_0 = \sqrt{E_b / \rho_b}, \quad (2)$$

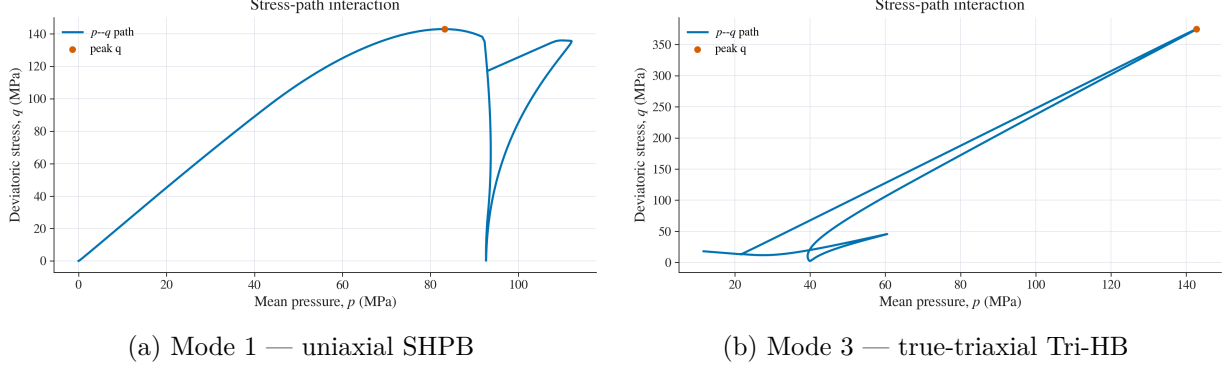


Figure 2: Mean-pressure–deviatoric-stress (p – q) trajectories for the uniaxial (a) and true-triaxial (b) configurations of the Tri-HB, plotted in *total* specimen stress $\sigma = \sigma_{\text{pre}} + \Delta\sigma_{\text{dyn}}$ (see text). Uniaxial loading (a) carries no pre-stress and traces a steep deviatoric path from the origin; the true-triaxial case (b), pre-stressed to (30, 20, 15) MPa, begins at its static point and populates the multiaxial p – q plane. The spread of accessible stress paths, set by the per-axis pre-stress, is the analytical advantage of the system.

whose general d’Alembert solution $u = f(x - C_0 t) + g(x + C_0 t)$ superposes forward- and backward-travelling parts. For a single travelling wave the axial strain and particle velocity are proportional, giving the Hopkinson identity

$$v(x, t) = \pm C_0 \varepsilon(x, t), \quad (3)$$

with the upper (lower) sign for a wave travelling in the $+x$ ($-x$) direction under the compression-positive convention above; a resistance gauge that measures strain therefore also measures the particle velocity through the bar wave speed. Summing the incident ($+x$) and reflected ($-x$) contributions then yields the face velocities of Eq. (5).

At a planar interface between media of acoustic impedance $Z = \rho c$, continuity of traction and velocity gives the stress reflection and transmission coefficients¹

$$R = \frac{\sigma_R}{\sigma_I} = \frac{Z_2 - Z_1}{Z_1 + Z_2}, \quad T = \frac{\sigma_T}{\sigma_I} = \frac{2Z_2}{Z_1 + Z_2}. \quad (4)$$

For a rock specimen softer than the steel bars ($Z_s < Z_b$) the reflection coefficient is negative, so the reflected pulse is tensile — the origin of the negative reflected strain ε_{re} recorded in compression tests, and a sign-diagnostic of the impedance match.

Equation (2) is exact only in the long-wavelength limit. The exact axisymmetric (Pochhammer–Chree) solution makes the phase velocity frequency-dependent, $c_\phi(ka) = C_0 \Phi(ka, \nu_b)$, where k is the wavenumber and a the bar radius: $\Phi \approx 1$ for $ka \lesssim 0.1$ but falls by a few percent as $ka \rightarrow 1$, so an unshaped rectangular pulse — rich in high- ka harmonics — develops spurious oscillations as it travels. Dispersion is negligible while the rise time τ_r satisfies $a/(c_\phi \tau_r) \lesssim 0.1$. The Tri-HB therefore uses a pulse-shaped, half-sine incident wave, which concentrates its energy at $ka \ll 1$ and is essentially dispersion-free for the bar geometries used here; the computational workspace represents the loading with a smooth half-sine incident wave for the same reason.

3.2 Force–velocity continuity and the three-wave method

At the incident-bar/specimen face the bar carries the superposed incident and reflected waves, while the transmission bar carries only the outgoing wave, so the two face forces and velocities are

$$F_1 = E_b A_b (\varepsilon_{\text{in}} + \varepsilon_{\text{re}}), \quad v_1 = C_0 (\varepsilon_{\text{in}} - \varepsilon_{\text{re}}); \quad F_2 = E_b A_b \varepsilon_{\text{tr}}, \quad v_2 = C_0 \varepsilon_{\text{tr}}. \quad (5)$$

¹For a 42CrMo steel bar $Z_b = \rho_b C_0 \approx 4.06 \times 10^7 \text{ kg m}^{-2} \text{ s}^{-1}$.

The average axial stress, strain rate and strain follow from (5). In general — valid at *all* times, equilibrium or not — the two-wave forms are

$$\sigma_x(t) = \frac{F_1 + F_2}{2A_s} = \frac{E_b A_b}{2A_s} (\varepsilon_{\text{in}} + \varepsilon_{\text{re}} + \varepsilon_{\text{tr}}), \quad (6)$$

$$\dot{\varepsilon}(t) = \frac{v_1 - v_2}{L_s} = \frac{C_0}{L_s} (\varepsilon_{\text{in}} - \varepsilon_{\text{re}} - \varepsilon_{\text{tr}}), \quad \varepsilon(t) = \int_0^t \dot{\varepsilon} d\tau, \quad (7)$$

where A_b, A_s are the bar and specimen cross-sections and L_s the specimen length. Only once the specimen has reverberated into dynamic stress equilibrium ($t \gtrsim 3 t_{\text{travel}}$, $t_{\text{travel}} = L_s/c_p$) do the two face forces equalise, $\varepsilon_{\text{in}} + \varepsilon_{\text{re}} = \varepsilon_{\text{tr}}$, and only then do (6)–(7) reduce to the familiar one-wave relations

$$\sigma_x(t) = \frac{E_b A_b}{A_s} \varepsilon_{\text{tr}}(t), \quad \dot{\varepsilon}(t) = -\frac{2C_0}{L_s} \varepsilon_{\text{re}}(t). \quad (8)$$

Agreement between the two-wave and one-wave forms is the reducer’s built-in equilibrium check; equilibrium itself is quantified by the force-balance ratio

$$\mathcal{R}(t) = \frac{|F_1(t) - F_2(t)|}{\frac{1}{2}(|F_1(t)| + |F_2(t)|)} < 0.05, \quad t_{\text{eq}} \approx (3\text{--}5) L_s/c_p, \quad (9)$$

the force magnitudes in the denominator keeping $\mathcal{R}$ well-defined through the early transient. The lateral response is obtained from the two output bars on each transverse axis. Each transverse axis is instrumented symmetrically with two output (transmission) bars that carry only outgoing waves, so each output-bar face force and velocity are $E_b A_b \varepsilon_{yk}$ and $C_0 \varepsilon_{yk}$ ($k = 1, 2$). Under the standard one-dimensional wave assumption with symmetric lateral loading — planar (rigid) specimen faces, no interface rotation and negligible transverse inertia — averaging the two opposing faces gives the lateral stress, while the rate at which the specimen widens against its lateral dimensions d_y, d_z gives the lateral strain rate,

$$\sigma_y = \frac{E_b A_b}{2A_s} (\varepsilon_{y1} + \varepsilon_{y2}), \quad \sigma_z = \frac{E_b A_b}{2A_s} (\varepsilon_{z1} + \varepsilon_{z2}), \quad (10)$$

$$\dot{\varepsilon}_y = -\frac{C_0}{d_y} (\varepsilon_{y1} + \varepsilon_{y2}), \quad \dot{\varepsilon}_z = -\frac{C_0}{d_z} (\varepsilon_{z1} + \varepsilon_{z2}), \quad (11)$$

with A_s here the loaded lateral area and $\varepsilon_y = \int_0^t \dot{\varepsilon}_y d\tau$. The minus sign reflects the convention: under axial compression both output bars recede as the specimen expands laterally, an outward (compression-negative, i.e. extensile) lateral strain. These lateral outputs are Poisson-driven — the X pulse compresses the specimen, which expands laterally and drives output waves outward along the Y and Z bars, recorded by gauges G3–G6 — and the Virtual Tri-HB performs the reduction simultaneously on all three axes, so a single test yields the full $\sigma_x, \sigma_y, \sigma_z$ history rather than one axial curve.

3.3 Validity of the three-wave method under static pre-stress

In the static–dynamic configuration the hydraulic cylinders pre-load the bars before the pulse arrives, so the bar ends are no longer free. Because the bar is linear-elastic, the total field superposes a uniform static part and the propagating dynamic part, $\sigma = \sigma_s + \sigma_d$ with $\sigma_s = F_{\text{pre}}/A_b = \text{const}$ ($d\sigma_s/dx = 0$). Substituting into (2), the static part drops out and the dynamic part obeys the same wave equation, so the three-wave relations apply unchanged to the dynamic *increments* $\varepsilon_{\text{in}}, \varepsilon_{\text{re}}, \varepsilon_{\text{tr}}$ — the pre-stress is merely a constant gauge offset, removed by baselining before the pulse. The superposition is valid provided the bar stays elastic at all times,

$$\sigma_{d,\text{peak}} < \sigma_{\text{prop}} - |\sigma_s|, \quad (12)$$

so the pre-stress consumes part of the dynamic-amplitude budget (the bar-plasticity limit the workspace enforces). The bar–specimen interfaces, being unilateral, must also stay in contact: a

reflected tensile excursion must not exceed the static pre-compression, $|\sigma_d| \leq \sigma_s$, or the contact separates and the boundary condition turns nonlinear — here the pre-stress *helps*, holding the interfaces closed. The far ends now carry a static force, but the reflection a pulse sees there is governed by the termination’s *dynamic* impedance, not by that force, and is in any case delayed beyond the measurement window by the bar length.

The constancy of the *bar* properties is what makes this robust. The bar wave speed is pre-stress-independent to within a sub-percent acoustoelastic shift, $\Delta C_0/C_0 \approx (\beta/E_b)\sigma_s \lesssim 0.3\%$ with $\beta = \mathcal{O}(1\text{--}5)$ for steel, so E_b, C_0, A_b may be treated as constants and the bar-based reduction holds at every confinement. The *specimen*, by contrast, is genuinely pre-stress-dependent: confinement closes microcracks and stiffens the rock, so $E_s = E_s(p)$ and $c_p = c_p(p)$ can vary by tens of percent over the first tens of megapascals. This is precisely the response the test measures, but it means any specimen-property-based output — the equilibrium window $t_{\text{eq}} \propto L_s/c_p$ and the energy partition — must use the *confined* values; and under true-triaxial pre-stress the closure is directional, rendering the rock elastically anisotropic ($c_{p,x} \neq c_{p,y} \neq c_{p,z}$), a stress-induced anisotropy the three orthogonal bars can resolve. This clean separation — a linear, pre-stress-independent bar reduction carrying a nonlinear, pre-stress-dependent specimen response — is what legitimises and gives meaning to static–dynamic coupled testing.

Finally, the sandwiched specimen itself does *not* superpose — it is the nonlinear element. The bars still record the true dynamic increments, but those increments are the rock’s response at its *pre-stressed* tangent, which is the coupling the test measures; the specimen’s absolute state is the static pre-load plus that increment, $\sigma_{\text{total}} = \sigma_{\text{pre}} + \Delta\sigma_{\text{dyn}}$ and $\varepsilon_{\text{total}} = \sigma_{\text{pre}}/E_s(p) + \Delta\varepsilon_{\text{dyn}}$, so only the strain reconstruction draws on the confined modulus $E_s(p)$ (the bar-based stress increment is self-contained). Confinement moreover *eases* the measurement: as crack closure raises $c_{p,s}$ and hence $Z_s = \rho_s c_{p,s}$, the equilibration time $t_{\text{eq}} \propto L_s/c_{p,s}$ shortens and the bar–specimen impedance mismatch $|R| = (Z_b - Z_s)/(Z_b + Z_s)$ falls, giving a cleaner transmitted signal (at the cost of a weaker reflected one).

4 The integrated computational workspace

The Virtual Tri-HB (`tri_hb_integrated.py`) is a single application that bundles six analysis steps under one shared experimental setup: (1) configuration and either simulation or reduction of bar-gauge data; (2) wave-propagation and equilibrium analysis; (3) stress-path and invariant analysis; (4) damage and energy analysis with directional descriptors; (5) a publication-ready summary; and (6) automatic generation of a report and presentation (L^AT_EX + PDF) from the current run. A separate page renders the live calibration/validation plan. Each step inherits the shared configuration, so the wave, stress-path and damage views remain mutually consistent. The stress-path stage resolves the total stresses into the invariants

$$p = \frac{1}{3}(\sigma_x + \sigma_y + \sigma_z), \quad q = \sqrt{3J_2}, \quad \cos 3\theta = \frac{3\sqrt{3}}{2} \frac{J_3}{J_2^{3/2}}, \quad (13)$$

where $s_{ij} = \sigma_{ij} - p\delta_{ij}$ is the deviatoric stress, $J_2 = \frac{1}{2}s_{ij}s_{ij} = \frac{1}{6}[(\sigma_x - \sigma_y)^2 + (\sigma_y - \sigma_z)^2 + (\sigma_z - \sigma_x)^2]$ and $J_3 = \det(s_{ij}) = s_1s_2s_3$ are its second and third invariants. With compression positive and the ordering $\sigma_1 \geq \sigma_2 \geq \sigma_3$, the Lode angle $\theta \in [0^\circ, 60^\circ]$ runs from the triaxial-compression meridian ($\theta = 0^\circ, \sigma_2 = \sigma_3$) to the triaxial-extension meridian ($\theta = 60^\circ, \sigma_1 = \sigma_2$), so that all three principal stresses — including the intermediate stress σ_2 through θ — enter the subsequent failure analysis (Fig. 3).

The time-coloured p – q trajectory (Fig. 3b) makes the loading history explicit and, like Fig. 2, is drawn in *total* stress $\sigma = \sigma_{\text{pre}} + \Delta\sigma_{\text{dyn}}$. Four stages are visible. (i) The path *begins at the static pre-stress point* (p_0, q_0) set by the true-triaxial confinement, not at the origin. (ii) As the dynamic X pulse arrives, the axial stress rises ahead of the Poisson-driven lateral outputs — which reach

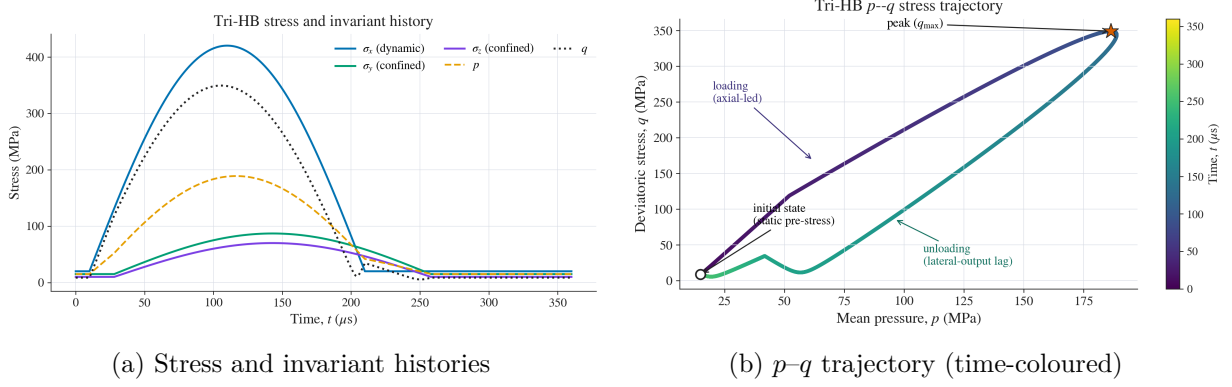


Figure 3: Stress-path analysis (Step 3) for the gas-gun Tri-HB (Mode 3): a dynamic X pulse superposed on a true-triaxial static pre-stress, with the Poisson-driven lateral output recorded on the Y and Z bars. (a) Total principal stresses with the mean (p) and deviatoric (q) invariants; (b) the resulting p - q trajectory (total stress, time-coloured), annotated with the initial static-pre-stress state, the axial-led loading branch, the deviatoric peak, and the lower unloading branch produced by the axial-lateral timing lag.

the Y, Z output bars only after a finite lateral transit — so the early path is *axial-led*: q climbs steeply along the upper (loading) branch with comparatively little change in p . (iii) At the pulse crest the trajectory reaches *peak* deviatoric stress $q_{\max}$, where the failure index $F = q/q_f$ is largest. (iv) As the pulse *unloads*, the axial stress relaxes before the lagged lateral outputs decay, so the return follows a distinct *lower* (unloading) branch and the trajectory does not retrace itself before settling back to the static point. The open loop between the two branches is therefore a *timing* effect — the phase lag between the prescribed axial pulse and the Poisson lateral outputs — and not a plastic or damage hysteresis: because the stresses are prescribed they return to the pre-stress state, while the irreversible response appears instead in the strain, stiffness and damage histories (§5.3–5.4). With independent stress and strain measured on each axis, the experiment will resolve whether the true material response adds genuine (dissipative) hysteresis to this kinematic loop.

5 Constitutive advances

5.1 True-triaxial, rate-dependent failure envelope

Failure is evaluated through the index $F = q/q_f$ against a pressure-dependent envelope with a deviatoric (Lode) shape and an optional rate dependence,

$$q_f = (A_{\text{eff}} + B_{\text{eff}} p^n) h(\theta), \quad A_{\text{eff}} = A \cdot \text{DIF}_c, \quad (14)$$

with the (compression) dynamic increase factor DIF_c defined in Eq. (16) below. The meridian parameters are not free. Requiring the envelope to reproduce the Step-1 strength law $\sigma_1 = (\text{UCS} + k_{\text{conf}} \sigma_3) \text{DIF}_c$ in triaxial compression ($\theta = 0$, $h = 1$, $n = 1$) and eliminating σ_3 through p and q yields the linear Mohr–Coulomb meridian

$$A = \frac{3 \text{UCS}}{k_{\text{conf}} + 2}, \quad B = \frac{3(k_{\text{conf}} - 1)}{k_{\text{conf}} + 2}, \quad n = 1, \quad (15)$$

so that $F = q/q_f = 1$ at the static failure point by construction (verified to machine precision for all confinement ratios k_{conf}). The exponent n is a fitting parameter for curved meridians; B_{eff} then carries units of MPa^{1-n} so that q_f remains a stress for any n , with $n = 1$ the default linear case. The deviatoric shape $h(\theta)$ admits three forms: a legacy cap, a linear extension-weakened interpolation, and a convex Willam–Warnke form (Willam and Warnke, 1975). Both new forms

use the meridian ratio ρ_t/ρ_c and make the extension meridian *weaker* than compression — the physically correct sense for brittle rock — so the intermediate principal stress effect enters with the right sign (Fig. 4a). The default is the Lode form with $\rho_t/\rho_c = 0.70$. The optional rate factor DIF_c inflates the envelope with strain rate (Fig. 4b) so that the failure *onset* shifts dynamically, complementing the rate sensitivity of the damage kinetics. Rather than a single semi-logarithmic slope (Zhao, 2000), we adopt the experimentally fitted, piecewise dynamic increase factor of Zhu et al. (2024) — a static log-linear branch below a transition strain rate and a power-law dynamic branch above it,

$$\text{DIF}_c = \begin{cases} k_c \log_{10} \dot{\epsilon} + b, & \dot{\epsilon} < 10 \text{ s}^{-1}, \\ k_c \log_{10} \dot{\epsilon} + b - \frac{10^\beta}{\alpha} + \frac{\dot{\epsilon}^\beta}{\alpha}, & \dot{\epsilon} \geq 10 \text{ s}^{-1}, \end{cases} \quad (16)$$

$$\text{DIF}_t = \begin{cases} k_t \log_{10} \dot{\epsilon} + b, & \dot{\epsilon} < 1 \text{ s}^{-1}, \\ k_t \log_{10} \dot{\epsilon} + b - \frac{1}{\alpha} + \frac{\dot{\epsilon}^{2\beta}}{\alpha}, & \dot{\epsilon} \geq 1 \text{ s}^{-1}, \end{cases} \quad (17)$$

where $\text{DIF}_c, \text{DIF}_t$ are the compression and tension factors, $\dot{\epsilon}$ is the strain rate (s^{-1}), k_c, k_t the static gradient coefficients, b a fitting constant, and $\alpha = 10, \beta = 0.57$; the inflection (transition) rates are 10 s^{-1} in compression and 1 s^{-1} in tension. Because the rate effect below the transition is small, the static branch is taken rate-independent ($k_c = k_t = 0, b = 1$), so $\text{DIF} = 1$ below the transition and rises as a power law above it (Fig. 4b); DIF_c scales the compressive envelope and DIF_t the tensile strength used in the directional damage of §5.4.

The Mohr–Coulomb meridian is adopted as the *default* not because rock is assumed insensitive to the intermediate principal stress, but because the surface is written in the standard invariant decomposition $q_f(p, \theta) = [\text{meridian}(p)] h(\theta)$, in which σ_2 enters through the Lode angle θ (Eq. (13)) — that is, through the deviatoric shape $h(\theta)$, not through the meridian. The meridian only fixes the compression reference ($\theta = 0, \sigma_2 = \sigma_3$), while the σ_2 -dependence is carried by $h(\theta)$, exactly as in the Willam–Warnke (Willam and Warnke, 1975) and related meridian–deviatoric criteria. The linear Mohr–Coulomb meridian is chosen for that reference because it is the minimal, identifiable form that reduces *exactly* to the routinely measured unconfined strength and confinement (Mode 2) series, makes the surface consistent with the Step-1 stress–strain model ($F = 1$ at peak), and needs no true-triaxial data to instantiate — a calibration anchor available *before* the full true-triaxial campaign, not a claim that σ_2 is unimportant.

For studies that target the intermediate-stress effect directly, this default can be *replaced* by a criterion that depends on σ_2 intrinsically. The workspace offers two such calibrated true-triaxial criteria — Mogi–Coulomb and modified Lade — which §5.2 details, compares with the default, and applies to the Tri-HB stress path; ultimately the meridian, the Lode ratio ρ_t/ρ_c and the criterion choice are all to be *fitted* to the Tri-HB’s own true-triaxial data, the measurement the system is built to provide.

5.2 True-triaxial criteria: Mogi–Coulomb and modified Lade

The one-parameter Lode shape is a first-order surrogate; for studies that target the intermediate principal stress the workspace offers two calibrated true-triaxial criteria, both derived from the *same* unconfined strength and confinement ratio so all three surfaces are directly comparable. The *Mogi–Coulomb* criterion (Mogi, 1971; Al-Ajmi and Zimmerman, 2005) sets the octahedral shear against the mean of the extreme principal stresses,

$$\tau_{\text{oct}} = a + b \sigma_{m,2}, \quad \sigma_{m,2} = \frac{1}{2}(\sigma_1 + \sigma_3), \quad \tau_{\text{oct}} = \frac{\sqrt{2}}{3} q \quad (\text{Eq. (13)}), \quad (18)$$

with $a = \frac{2\sqrt{2}}{3}c \cos \phi$ and $b = \frac{2\sqrt{2}}{3} \sin \phi$ from the Coulomb cohesion and friction angle implied by the strength data, $\sin \phi = (k_{\text{conf}} - 1)/(k_{\text{conf}} + 1)$ and $c = \text{UCS} (1 - \sin \phi)/(2 \cos \phi)$. Although $\sigma_{m,2}$

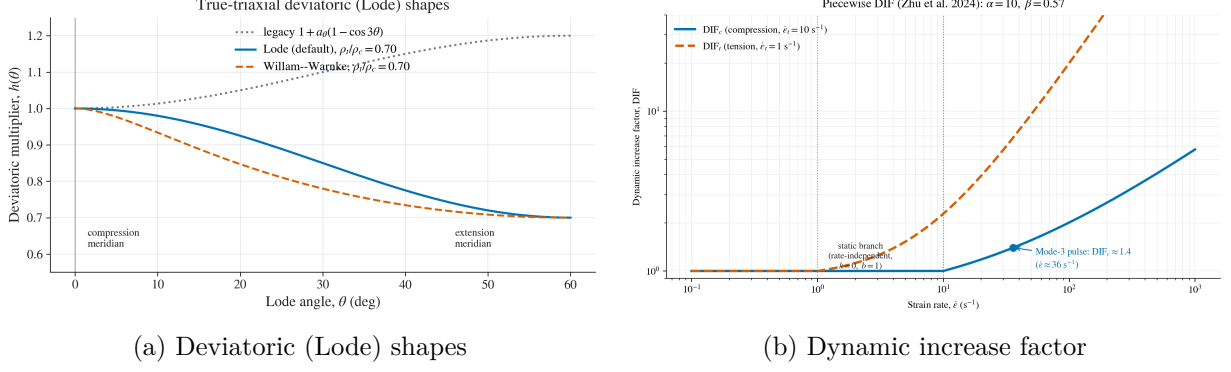


Figure 4: True-triaxial and rate-dependent failure envelope. (a) The default Lode and Willam-Warnke shapes correctly weaken the extension meridian relative to compression ($\rho_t/\rho_c = 0.70$), capturing the σ_2 effect; the legacy cap (dotted) has the opposite, unphysical sense. (b) Piecewise dynamic increase factor of Zhu et al. (2024) (Eqs. (16)–(17)): a rate-independent static branch ($\text{DIF} = 1$) below the transition rate (10 s^{-1} in compression, 1 s^{-1} in tension) and a power-law dynamic branch above it ($\alpha = 10$, $\beta = 0.57$); DIF_t rises faster ($\propto \dot{\epsilon}^{2\beta}$) than DIF_c ($\propto \dot{\epsilon}^\beta$). The Mode-3 pulse ($\dot{\epsilon} \approx 36 \text{ s}^{-1}$) gives $\text{DIF}_c \approx 1.4$.

omits σ_2 , the octahedral shear τ_{oct} retains it, so the criterion is σ_2 -dependent. The *modified Lade* criterion (Ewy, 1999) uses the smooth, corner-free form

$$\frac{I_1''^3}{I_3''} = 27 + \eta, \quad I_1'' = \sum_i (\sigma_i + S), \quad I_3'' = \prod_i (\sigma_i + S), \quad S = \frac{c}{\tan \phi}, \quad \eta = \frac{4 \tan^2 \phi (9 - 7 \sin \phi)}{1 - \sin \phi}, \quad (19)$$

the translation S shifting the apex to embed the tensile strength. By construction both reduce to $\sigma_1 = \text{UCS}$ when unconfined and to the Coulomb meridian $\sigma_1 = \text{UCS} + k_{\text{conf}}\sigma_3$ in triaxial compression ($\sigma_2 = \sigma_3$). Two further σ_2 -sensitive surfaces are provided. *Yu's unified strength theory* (Yu, 2002) is the piecewise twin-shear form

$$\max \left[\sigma_1 - \frac{k_{\text{conf}}}{1+b} (b\sigma_2 + \sigma_3), \quad \frac{1}{1+b} (\sigma_1 + b\sigma_2) - k_{\text{conf}}\sigma_3 \right] = \text{UCS}, \quad (20)$$

whose unified parameter $b \in [0, 1]$ interpolates from Mohr–Coulomb ($b = 0$) to the twin-shear bound ($b = 1$); the two linear branches switch at $\sigma_2 = (\sigma_1 + k_{\text{conf}}\sigma_3)/(k_{\text{conf}} + 1)$, producing the σ_2 enhancement. The *modified Wiebols–Cook* criterion (Wiebols and Cook, 1968) is the smooth quadratic

$$\sqrt{J_2} = A_w + B_w J_1 + C_w J_1^2, \quad J_1 = \sigma_1 + \sigma_2 + \sigma_3, \quad (21)$$

with A_w, B_w, C_w fixed by the unconfined strength and the triaxial-compression meridian; here the σ_2 -dependence enters through J_1 and J_2 . Polyaxial benchmarking finds such σ_2 -sensitive criteria outperform σ_2 -independent ones for rock (Colmenares and Zoback, 2002).

The five surfaces differ in *how* strongly the intermediate stress strengthens the rock. To reflect a realistic test, Figure 5a uses the *dynamic* strength — the quasi-static surface inflated by a representative rate factor $\text{DIF}_c \approx 1.4$ (Eq. (16) at $\dot{\epsilon} \approx 36 \text{ s}^{-1}$), consistent with the multi-impact tests (Wang, Zhao, et al., 2022) — and treats the principal stresses as *total*. The static pre-stress is capped at $\sim 50 \text{ MPa}$ by the hydraulic confinement, so the intermediate stress can only exceed 50 MPa through the *dynamic confinement* the impact develops on the lateral bars ($\sigma_2, \sigma_3 = \sigma_{\text{pre}} + \Delta\sigma_{\text{dyn}}$); this is why the dynamic strengths σ_1 reach several hundred MPa rather than the quasi-static values. At $\sigma_3 = 50 \text{ MPa}$ the classical (σ_2 -blind) Mohr–Coulomb strength is flat at $\sigma_1 = \text{UCS} \cdot \text{DIF} + k_{\text{conf}}\sigma_3$, whereas every σ_2 -aware surface rises with σ_2 across the physically accessible range — the intermediate-stress strengthening (Mogi, 1971) — the five spreading by ~ 15 – 20% at the upper end and all coinciding on the triaxial-compression ($\sigma_2 = \sigma_3$) Coulomb line by construction. Evaluated along the Tri-HB Mode-3 stress path of Fig. 3 (Fig. 5b), the criteria

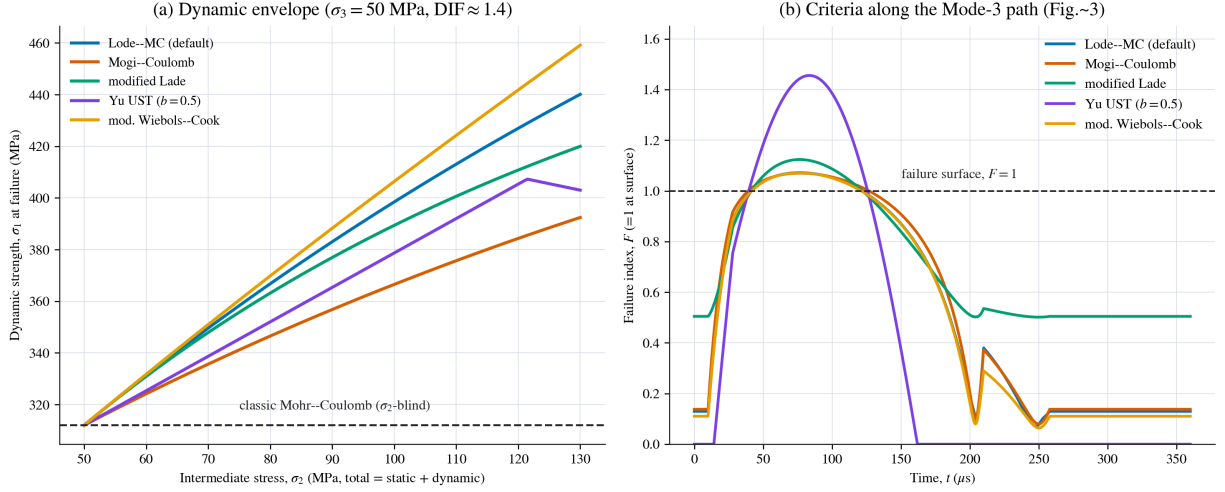


Figure 5: True-triaxial failure criteria at realistic Tri-HB conditions, all calibrated to the same dynamic strength (UCS·DIF, $DIF \approx 1.4$) and $k_{\text{conf}} = 4$. (a) Failure strength σ_1 versus the *total* intermediate stress σ_2 at $\sigma_3 = 50$ MPa: the classical Mohr–Coulomb surface is σ_2 -blind (flat), while the Lode–MC, Mogi–Coulomb, modified-Lade, Yu-UST and modified Wiebols–Cook surfaces capture the intermediate-stress strengthening; σ_2 exceeds the 50 MPa static cap only through dynamic confinement. (b) The corresponding failure indices ($F = 1$ at the surface) along the Tri-HB Mode-3 stress path of Fig. 3.

report different peak failure indices for the *same* loading ($F_{\text{max}} \approx 1.1\text{--}1.5$), so the choice of surface materially changes the predicted dynamic strength. Which surface is correct for a given rock is an experimental question — precisely the σ_2 -resolved measurement the Tri-HB provides — after which the meridian, the Lode ratio and the criterion are fitted to the data.

5.3 Energy-consistent damage dissipation

Damage D evolves by a Perzyna-type overstress law with load-shedding feedback,

$$\dot{D} = \frac{(1-D)^\alpha}{\tau_D} \left\langle \frac{F_{\text{eff}} - 1}{F_0} \right\rangle^m \left(\frac{|\dot{\epsilon}_{\text{eq}}|}{\dot{\epsilon}_0} \right)^\beta, \quad F_{\text{eff}} = (1-D) F, \quad (22)$$

where $\langle \cdot \rangle$ is the Macaulay bracket and $\tau_D, \alpha, m, \beta, F_0, \dot{\epsilon}_0$ are kinetic constants. The load-shedding factor $F_{\text{eff}} = (1-D) F$ is the key feature: as D grows, the effective overstress $\langle F_{\text{eff}} - 1 \rangle$ shrinks and the growth rate decays, so D approaches a finite steady value rather than diverging to unity (Perzyna, 1966; Lemaitre, 1985). The energy partition is, moreover, now derived from this single constitutive relation. Damage degrades the secant modulus to $E_0(1-D)$, so the specimen carries the damage-coupled elastic strain and the work, recoverable energy and dissipation become

$$\varepsilon_i = \frac{\sigma_i - \nu(\sigma_j + \sigma_k)}{E_0(1-D)}, \quad W_{\text{input}} = \int_0^t \sigma_i \dot{\varepsilon}_i d\tau, \quad W_{\text{el}} = \frac{1}{2} \sigma_i \varepsilon_i, \quad W_{\text{diss}} = \langle W_{\text{input}} - W_{\text{el}} \rangle. \quad (23)$$

The first law then closes by construction, $W_{\text{input}} = W_{\text{el}} + W_{\text{diss}}$: the input work ends above zero by exactly the dissipated part, the recoverable energy returns toward zero on unloading, and the dissipation rises monotonically (Fig. 6). This removes the inconsistency of treating the input work as fully recoverable while estimating dissipation from a separate integral. The partition is non-negative by construction, $W_{\text{diss}} = \langle W_{\text{input}} - W_{\text{el}} \rangle \geq 0$, and monotone in all reference cases (Fig. 6); we therefore call it *energetically self-consistent* rather than claiming full thermodynamic admissibility, for which a pointwise dissipation inequality $\dot{W}_{\text{diss}} \geq 0$ from a Clausius–Duhem argument with the free energy remains future work. Equation (23) is, furthermore, the *scalar*-damage partition; the orthotropic model of §5.4 is a first-stage extension that still inherits it

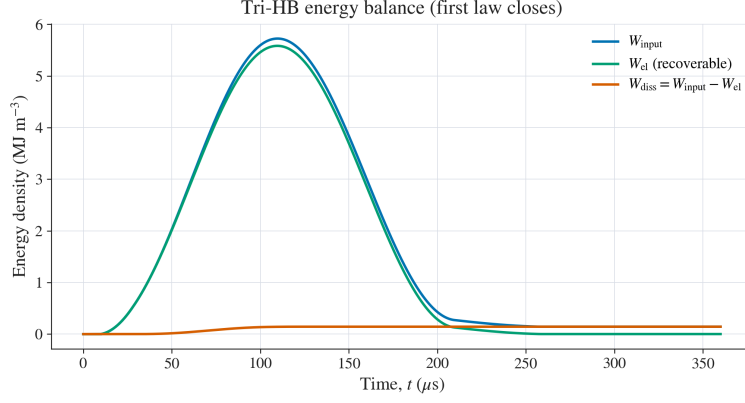


Figure 6: Energy-density partition (Step 4) for the gas-gun Tri-HB case. With the damage-coupled strain the balance closes exactly: $W_{\text{input}} = W_{\text{el}} + W_{\text{diss}}$, with W_{diss} the non-recoverable (dissipated) energy.

through $D = \max_i D_i$, with the fully anisotropic form — W_{el} built from a damaged compliance tensor $\mathbf{S}(\mathbf{D})$ rather than a scalar secant modulus — also left to future work.

5.4 Orthotropic damage for blasting-induced fracturing

Rock blasting produces *directional* fracture: radial cracks and spall open perpendicular to the local tension, so the stiffness and wave-speed loss differ by direction. A scalar D cannot represent this. The Tri-HB computational workspace therefore offers a diagonal damage tensor $\mathbf{D} = \text{diag}(D_x, D_y, D_z)$ aligned with the bars. Each component evolves through the same overstress kinetics, driven by a *dominant directional-tensile* term plus a *small shear/compression* term,

$$\dot{D}_i = \frac{(1 - D_i)^\alpha}{\tau_D} \left\langle \frac{(1 - D_i)F_i - 1}{F_0} \right\rangle^m \left(\frac{|\dot{\varepsilon}_{\text{eq}}|}{\dot{\varepsilon}_0} \right)^\beta, \quad F_i = \frac{\langle -\varepsilon_i \rangle_+}{\varepsilon_{t0}} + \kappa_s \left\langle \frac{F - F_{s0}}{1 - F_{s0}} \right\rangle, \quad (24)$$

for $i = x, y, z$, with $\varepsilon_{t0} = \sigma_t/E_0$ (σ_t the tensile strength), $F = q/q_f$ the deviatoric failure index of §5.1, and $(\kappa_s, F_{s0}) = (1, 0.5)$ here. The first term is *tensile*: in the compression-positive convention a tensile strain is negative, so $\langle -\varepsilon_i \rangle_+ = \max(-\varepsilon_i, 0)$ extracts the extensile strain that opens cracks on the plane normal to axis i . The second term is an *isotropic* shear/compression contribution that switches on once the deviatoric index exceeds F_{s0} ; it represents the wing-crack, shear-microcrack and grain-crushing damage that real rock accrues *even under net compression*. Because F only modestly exceeds unity at the dynamic peak ($F_{\text{max}} \approx 1.14$ here), this term accrues only a small axial damage, whereas the tensile term — fed by the large lateral extensile strains — dominates the lateral axes. Axial X compression therefore drives large lateral components D_y, D_z (axial splitting) while giving a *small but nonzero* axial component $D_x \ll D_y, D_z$ (here $D_x \approx 0.1$ versus $D_y \approx 0.75$, $D_z \approx 0.83$) — consistent with the modest axial-stiffness loss seen in dynamically compressed rock, rather than the unphysical $D_x = 0$ of a pure-tension model. Confinement on an axis suppresses its tensile damage, so the least-confined lateral axis (Z) is the most damaged (Fig. 7a). The per-axis modulus degrades independently, $E_i = E_0(1 - D_i)$ (Fig. 7b), as does the directional wave speed $c_{p,i} = c_{p0}\sqrt{1 - D_i}$. This formulation is rooted in the microcrack-damage tradition for brittle rock (Grady and Kipp, 1980; Taylor et al., 1986) but resolves damage by direction. Decisively, because the three orthogonal bars measure direction-resolved modulus and wave-speed loss, each D_i can be calibrated *component-by-component* from a single true-triaxial test — a measurement a uniaxial SHPB cannot provide.

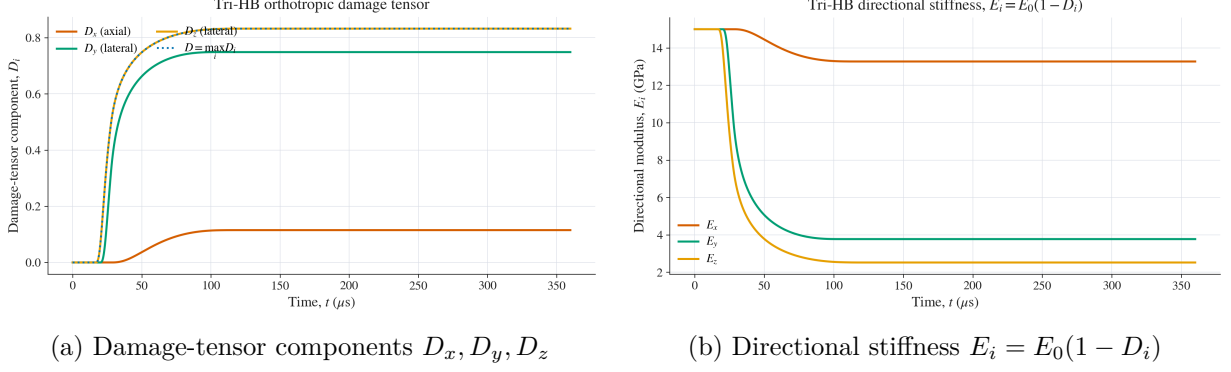


Figure 7: Orthotropic (Stage-1) damage for the gas-gun Tri-HB (Mode 3) case. (a) The diagonal damage-tensor components: a dominant tensile driver gives large lateral damage ($D_z > D_y$, the least-confined axis most damaged), while a small shear/compaction driver gives a nonzero but much smaller axial $D_x \approx 0.1$; the scalar $D = \max_i D_i$ governs the energy/stiffness summary. (b) The resulting directional modulus loss: E_x falls modestly while E_y, E_z fall sharply — the per-axis quantity the three bar pairs measure.

5.5 Cyclic dynamic loading and cumulative damage

Because each axis is loaded through bars rather than a sealed chamber, the Tri-HB can hold the specimen at its in-situ static pre-stress, fire one dynamic pulse, reset to the same static state, and re-fire the *same* pulse — repeating until the rock fails. This reproduces repeated blasting, repeated rockburst loading and dynamic fatigue of a pre-stressed rock mass (Wang, Zhao, et al., 2022), none of which a single-shot SHPB can access. At the high amplitudes used in practice the rock fails within a few impacts (typically $N_f \lesssim 10$); tens or hundreds of impacts are neither feasible nor observed.

Damage carries over between impacts and degrades the strength $q_f(D) = q_{f,0}(1 - D)^{c_d}$, so an impact the rock survives once accumulates to failure (Zhu et al., 2024). With impact index k and the failure index $F = q/q_f$ from the active criterion (Lode, Mogi–Coulomb or modified Lade), the cumulative-damage map is

$$\begin{aligned} D_k &= \min(D_{k-1} + \Delta D_k, 1), & F_k &= F_1(1 - D_{k-1})^{-c_d}, \\ \Delta D_k &= K \langle (1 - D_{k-1})F_k - F_{th} \rangle^m (1 - D_{k-1})^\alpha, \end{aligned} \quad (25)$$

which keeps the same Perzyna kinetics (Perzyna, 1966; Lemaitre, 1985) integrated over one pulse — appropriate because repeated impact is *low-cycle* ($N \lesssim 20$) fatigue, in which each high-amplitude impact is a genuine overstress excursion. The effective driving $(1 - D)F_k = F_1(1 - D)^{1-c_d}$ exposes three regimes through the exponent $1 - c_d$: shakedown ($c_d < 1$), linear Palmgren–Miner accumulation ($c_d = 1$), and accelerating progressive failure ($c_d > 1$). The last is *abrupt*: the strength-degradation feedback produces a sharp sigmoid that cascades to failure once the degraded strength is first exceeded, at the critical damage $D^* = 1 - F_1^{1/c_d}$ — reproducing the sudden damage threshold seen in synchrotron micro-CT of multi-impact tests (Wang, Zhao, et al., 2022) (Fig. 8).

Equation (25) is the spatially uniform reduction of a two-scale field problem. The fast intra-impact wave IBVP couples the wave equation to the local damage rate,

$$\rho \partial_{tt} u = \partial_x [E_0(1 - D) \partial_x u], \quad \partial_t D = \frac{(1 - D)^\alpha}{\tau_D} \left\langle \frac{(1 - D)^{F-1}}{F_0} \right\rangle^m \left(\frac{|\dot{\epsilon}_{eq}|}{\dot{\epsilon}_0} \right)^\beta, \quad (26)$$

with the *inherited* damage field $D(x, 0) = D_{k-1}(x)$ as initial condition, so the bar coefficients $E_0(1 - D)$, c_p and Z become evolving heterogeneous fields. Averaging over one pulse gives the slow inter-impact drift $\partial_N D = \int_0^{T_p} \partial_t D \, d\tau$; because the degraded strength softens, the local form

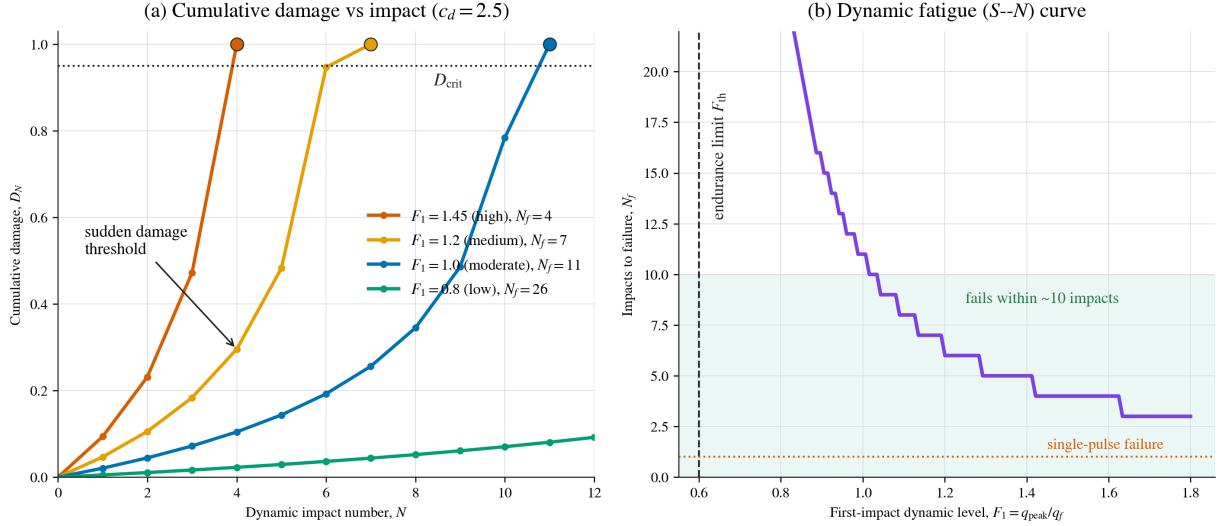


Figure 8: Cyclic (repeated) dynamic loading. (a) Cumulative damage versus impact number: high levels fail the rock within a few impacts through a sudden threshold (sharp sigmoid), low levels need impractically many. (b) The dynamic-fatigue ($S-N$) curve — impacts-to-failure fall into the realistic $\lesssim 10$ band at high level and diverge at the endurance limit F_{th} .

is ill-posed and is regularised by a gradient (nonlocal) term with internal length ℓ (Pijaudier-Cabot and Bažant, 1987),

$$\frac{\partial D}{\partial N} = \frac{1}{\tau_N} \langle (1 - D) \bar{F} - F_{th} \rangle^m (1 - D)^\alpha, \quad \bar{D} = D - \ell^2 \nabla^2 D, \quad (27)$$

with $\nabla D \cdot \mathbf{n} = 0$ on $\partial\Omega$; ℓ sets the fracture-band width. The per-impact damage is independently measurable: the soft-recovered specimen is micro-CT scanned after each impact, and the directional crack-density tensor gives $D_i = 1 - E_i/E_0$ to compare against the predicted D_k , so ℓ , τ_N and c_d are calibrated from the measured damage-versus-impact trajectory and the locus of per-impact peak stresses recovers the monotonic post-peak envelope.

5.6 Spatial damage verification: gradient field and phase-field

The two-scale formulation of §5.5 (Eqs. (26)–(27)) already carries the gradient regularisation term $\bar{D} = D - \ell^2 \nabla^2 D$; the present section makes the spatial field explicit, extends it to a thermodynamically consistent phase-field form, and calibrates both models from the CT scans, bar-signal records, fracture-area measurements and high-speed-camera data of the Tri-HB campaign.

Level 3 — gradient-regularised damage field. Lifting D from the spatially uniform scalar of §5.5 to a field $D(\mathbf{x}, t)$ over the specimen domain Ω gives the two-field coupled problem

$$\rho \ddot{\mathbf{u}} = \nabla \cdot [(1 - D) \mathbf{E}_0 : \nabla^s \mathbf{u}], \quad D - \ell^2 \nabla^2 D = D_{local}(\boldsymbol{\varepsilon}(\mathbf{x}, t)), \quad (28)$$

with the natural boundary condition $\nabla D \cdot \mathbf{n} = 0$ on $\partial\Omega$; bar signals supply the kinematic boundary conditions for the momentum equation. The Helmholtz operator smears the pointwise Perzyna damage D_{local} over a band of width $\approx 2\sqrt{2}\ell$ (Fig. 9a), removing the ill-posed localisation of the local form ($\ell = 0$) and making the solution mesh-independent (Pijaudier-Cabot and Bažant, 1987). The internal length is calibrated directly from the CT crack spacing δ_{crack} measured after each Tri-HB impact, $\ell \approx \delta_{crack}/\sqrt{2}$; the computed $D(\mathbf{x}, k)$ at cycle k is then compared slice-by-slice against the CT attenuation contrast.

Level 4 — phase-field fracture. The phase-field approach replaces D with a smooth field $\phi(\mathbf{x}, t) \in [0, 1]$ ($\phi = 0$ intact, $\phi = 1$ broken) derived from the single regularised energy

functional (Bourdin et al., 2000)

$$\mathcal{E}(\mathbf{u}, \phi) = \int_{\Omega} \left[(1 - \phi)^2 \Psi^+(\boldsymbol{\varepsilon}) + \Psi^-(\boldsymbol{\varepsilon}) + G_c \left(\frac{\phi^2}{2\ell} + \frac{\ell}{2} |\nabla \phi|^2 \right) \right] d\Omega, \quad (29)$$

where $\Psi^\pm$ is the spectral tension/compression split of the elastic energy density (only the tensile part Ψ^+ drives fracture, compression is undamaged), and G_c [J m⁻²] is the Griffith fracture energy. Stationarity $\delta\mathcal{E} = 0$ yields, alongside the damaged momentum equation, the phase-field PDE with viscous regularisation

$$\eta \dot{\phi} - G_c \ell \nabla^2 \phi + \frac{G_c}{\ell} \phi = 2(1 - \phi) \Psi^+(\boldsymbol{\varepsilon}), \quad (30)$$

where η [Pa s m] bounds the crack propagation speed (calibrated from high-speed-camera measurements of the crack front during the Tri-HB pulse). The model is thermodynamically consistent by construction: $\delta = G_c \partial_t \Gamma_\ell(\phi) \geq 0$ for any $\dot{\phi} \geq 0$. The irreversibility constraint $\dot{\phi} \geq 0$ is enforced via the history field $H(\mathbf{x}, t) = \max_{s \leq t} \Psi^+(\mathbf{x}, s)$, which replaces Ψ^+ in the driving term.

The one-dimensional equilibrium profile of a single crack centred at x_0 is analytical,

$$\phi(x) = \exp\left(-\frac{|x - x_0|}{\ell}\right), \quad (31)$$

so the crack band width at the e^{-1} level equals 2ℓ , directly measurable from CT cross-sections (Fig. 9b). The correspondences $\phi^2 \leftrightarrow D$ (degradation function) and $\Psi^+ \leftrightarrow Y$ (thermodynamic driving force) link Level 4 to the scalar Level 1 closure of §5.3, so all four levels share the same internal length and the same Clausius–Duhem admissibility argument. The Griffith energy is estimated from the measured dissipation and CT fracture area,

$$G_c \approx \frac{\chi_f W_{\text{diss}}^{\text{total}}}{A_{\text{fracture}}}, \quad (32)$$

where $W_{\text{diss}}^{\text{total}}$ is the integrated dissipation from the Step-3 energy partition (Fig. 6), A_{fracture} is the total crack surface area from marching-cubes segmentation of the CT stack, and $\chi_f \in (0, 1)$ is the fraction of measured dissipation attributable to fracture surface creation (the remainder being micro-crack friction and heat; χ_f can be recovered independently from acoustic-emission energy partitioning or from the K_{Ic} -based reference $G_c = K_{\text{Ic}}^2/E_0$).

6 Validation and calibration roadmap

The advances above are constitutive and computational; the system is presented here as a measurement-and-modelling framework rather than a validated predictive model. The decisive step is experimental: a staged, hierarchical calibration that fixes the elastic and strength parameters before the damage kinetics, on a *calibration set* of tests, followed by *blind prediction* on a disjoint *validation set*. The Tri-HB’s unique strength is that its three independent axes allow each anisotropic-damage component, each meridian of the failure surface and each rate level to be probed directly. A practical matrix calibrates the meridian and Lode parameters on a subset of confinements and strain rates and blind-predicts held-out true-triaxial stress states ($\sigma_2 \neq \sigma_3$), while the directional damage components D_x, D_y, D_z are recovered post-test from per-axis ultrasonic modulus loss, $D_i = 1 - E_{i,\text{post}}/E_0$. Demonstrating that one calibrated parameter set predicts strength and directional fracture on a held-out loading path is what converts the framework from phenomenological to predictive, and is the natural subject of the experimental campaign now enabled by the system.

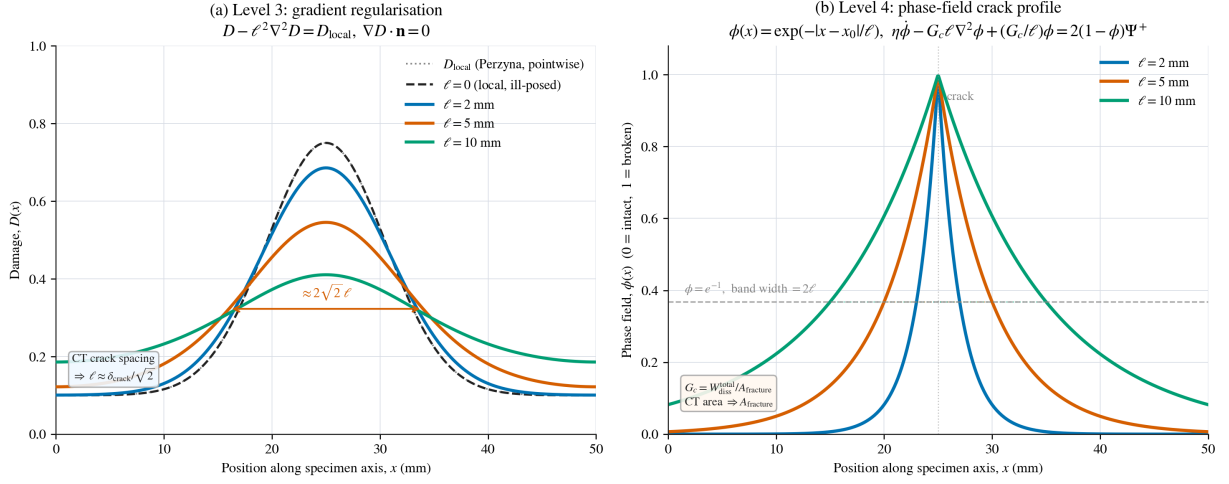


Figure 9: Spatial damage verification for the Mode-3 Tri-HB reference case (specimen length $L = 50$ mm, peak orthotropic damage $D_y \approx 0.75$ from Fig. 7). (a) Gradient regularisation (Eq. (28)): the pointwise Perzyna field $D_{\text{local}}(x)$ (dotted) localises sharply for $\ell = 0$ (dashed) and is smoothed into a finite process zone of width $\approx 2\sqrt{2}\ell$ as ℓ increases; ℓ is calibrated from the CT crack spacing δ_{crack} . (b) Phase-field crack profile (Eq. (31)): the equilibrium profile $\phi = \exp(-|x - x_0|/\ell)$ decays from unity at the crack centre; the band width 2ℓ at the e^{-1} level is directly measurable from CT cross-sections, and the Griffith energy follows from Eq. (32) using the Step-3 dissipation, the CT fracture area, and an acoustic-emission or K_{IC} -based estimate of χ_f . Both models share the same ℓ .

7 Conclusions

We have presented the Monash true-triaxial Hopkinson bar and its integrated computational workspace. The six principal advances are: (i) a unified treatment of uniaxial, axisymmetric-confined and true-triaxial static–dynamic loading as special cases of a single gas-gun system; (ii) a true-triaxial, Lode-angle- and rate-dependent failure envelope that captures the intermediate principal stress effect with the correct sign; (iii) an energy-consistent damage–dissipation partition in which the first law closes by construction with non-negative dissipation; (iv) an orthotropic damage tensor whose directional components reproduce the oriented fracture of rock blasting and can be calibrated component-by-component from the system’s direction-resolved measurements; (v) a cumulative-damage model for repeated (cyclic) dynamic loading that captures the sudden, few-impact dynamic-fatigue failure of a pre-stressed rock mass; and (vi) a spatial thermodynamic extension — a gradient-regularised damage field (internal length ℓ from CT crack spacing, Eq. (28)) and a phase-field formulation (Griffith energy G_c from the measured dissipation and CT fracture area, Eq. (32); crack profile validated against CT attenuation maps, Eq. (31)) that is thermodynamically consistent by construction and directly comparable to post-impact CT cross-sections from the Tri-HB campaign. Together, the hardware and its computational workspace establish a platform for quantitative, multiaxial dynamic constitutive modelling of rock; the immediate next step is the experimental calibrate-then-blind-predict campaign that the system uniquely enables.

Declarations

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Data and code availability. The Virtual Tri-HB workspace (`tri_hb_integrated.py`) and

the figure-generation scripts are part of the Tri-HB repository. Figures 1–8 are produced from prescribed-pulse reference cases by the workspace’s analysis pipeline. Figure 9 (§5.6) demonstrates the spatial framework on the same reference case; CT scan images, bar-signal records, fracture surface area measurements, and high-speed-camera data from the Tri-HB campaign are available from the corresponding author on reasonable request.

Author contributions. Q.B.Z. conceived the system and the workspace framework, implemented the analysis pipeline, and wrote the manuscript.

References

- Al-Ajmi, A. M. and R. W. Zimmerman (2005). “Relation between the Mogi and the Coulomb failure criteria”. In: *International Journal of Rock Mechanics and Mining Sciences* 42.3, pp. 431–439. DOI: 10.1016/j.ijrmms.2004.11.004.
- Bourdin, B., G. A. Francfort, and J.-J. Marigo (2000). “Numerical experiments in revisited brittle fracture”. In: *Journal of the Mechanics and Physics of Solids* 48.4, pp. 797–826. DOI: 10.1016/S0022-5096(99)00028-9.
- Christensen, R. J., S. R. Swanson, and W. S. Brown (1972). “Split-Hopkinson-bar tests on rock under confining pressure”. In: *Experimental Mechanics* 12.11, pp. 508–513. DOI: 10.1007/BF02320747.
- Colmenares, L. B. and M. D. Zoback (2002). “A statistical evaluation of intact rock failure criteria constrained by polyaxial test data for five different rocks”. In: *International Journal of Rock Mechanics and Mining Sciences* 39.6, pp. 695–729. DOI: 10.1016/S1365-1609(02)00048-5.
- Davies, R. M. (1948). “A critical study of the Hopkinson pressure bar”. In: *Philosophical Transactions of the Royal Society A* 240.821, pp. 375–457. DOI: 10.1098/rsta.1948.0001.
- Ewy, R. T. (1999). “Wellbore-stability predictions by use of a modified Lade criterion”. In: *SPE Drilling & Completion* 14.2, pp. 85–91. DOI: 10.2118/56862-PA.
- Frew, D. J., M. J. Forrestal, and W. Chen (2002). “Pulse shaping techniques for testing brittle materials with a split Hopkinson pressure bar”. In: *Experimental Mechanics* 42.1, pp. 93–106. DOI: 10.1007/BF02411056.
- Grady, D. E. and M. E. Kipp (1980). “Continuum modelling of explosive fracture in oil shale”. In: *International Journal of Rock Mechanics and Mining Sciences* 17.3, pp. 147–157. DOI: 10.1016/0148-9062(80)91361-3.
- Hopkinson, B. (1914). “A method of measuring the pressure produced in the detonation of high explosives or by the impact of bullets”. In: *Philosophical Transactions of the Royal Society A* 213.497–508, pp. 437–456. DOI: 10.1098/rsta.1914.0010.
- Kolsky, H. (1949). “An investigation of the mechanical properties of materials at very high rates of loading”. In: *Proceedings of the Physical Society. Section B* 62.11, pp. 676–700. DOI: 10.1088/0370-1301/62/11/302.
- Lemaitre, J. (1985). “A continuous damage mechanics model for ductile fracture”. In: *Journal of Engineering Materials and Technology* 107.1, pp. 83–89. DOI: 10.1115/1.3225775.
- Lindholm, U. S., L. M. Yeakley, and A. Nagy (1974). “The dynamic strength and fracture properties of Dresser basalt”. In: *International Journal of Rock Mechanics and Mining Sciences & Geomechanics Abstracts* 11.5, pp. 181–191. DOI: 10.1016/0148-9062(74)90885-7.
- Mogi, K. (1971). “Fracture and flow of rocks under high triaxial compression”. In: *Journal of Geophysical Research* 76.5, pp. 1255–1269. DOI: 10.1029/JB076i005p01255.
- Perzyna, P. (1966). “Fundamental problems in viscoplasticity”. In: *Advances in Applied Mechanics* 9, pp. 243–377. DOI: 10.1016/S0065-2156(08)70009-7.
- Pijaudier-Cabot, G. and Z. P. Bažant (1987). “Nonlocal damage theory”. In: *Journal of Engineering Mechanics* 113.10, pp. 1512–1533. DOI: 10.1061/(ASCE)0733-9399(1987)113:10(1512).
- Taylor, L. M., E. P. Chen, and J. S. Kuszmaul (1986). “Microcrack-induced damage accumulation in brittle rock under dynamic loading”. In: *Computer Methods in Applied Mechanics and Engineering* 55.3, pp. 301–320. DOI: 10.1016/0045-7825(86)90057-5.
- Wang, H. C. and Q. B. Zhang (2026). “Experimental testing, characterisation methods, and mechanical behaviour of blast-induced rock damage from uniaxial strain to multiaxial stress states”. In: *International Journal of Rock Mechanics and Mining Sciences* 201, p. 106436. DOI: 10.1016/j.ijrmms.2026.106436.
- Wang, H. C., J. Zhao, J. Li, C. H. Braithwaite, and Q. B. Zhang (2022). “Progressive fracturing of concrete under biaxial confinement and repetitive dynamic loadings: From damage to catastrophic failure”. In: *International Journal of Impact Engineering* 165, p. 104232. DOI: 10.1016/j.ijimpeng.2022.104232.

- Wiebols, G. A. and N. G. W. Cook (1968). “An energy criterion for the strength of rock in polyaxial compression”. In: *International Journal of Rock Mechanics and Mining Sciences & Geomechanics Abstracts* 5.6, pp. 529–549. DOI: 10.1016/0148-9062(68)90040-5.
- Willam, K. J. and E. P. Warnke (1975). “Constitutive model for the triaxial behaviour of concrete”. In: *IABSE Seminar on Concrete Structures Subjected to Triaxial Stresses*. Vol. 19. IABSE, pp. 1–30.
- Yu, M. H. (2002). “Advances in strength theories for materials under complex stress state in the 20th century”. In: *Applied Mechanics Reviews* 55.3, pp. 169–218. DOI: 10.1115/1.1472455.
- Zhang, Q. B. and J. Zhao (2014). “A review of dynamic experimental techniques and mechanical behaviour of rock materials”. In: *Rock Mechanics and Rock Engineering* 47.4, pp. 1411–1478. DOI: 10.1007/s00603-013-0463-y.
- Zhao, J. (2000). “Applicability of Mohr–Coulomb and Hoek–Brown strength criteria to the dynamic strength of brittle rock”. In: *International Journal of Rock Mechanics and Mining Sciences* 37.7, pp. 1115–1121. DOI: 10.1016/S1365-1609(00)00049-6.
- Zhou, Y. X., K. Xia, X. B. Li, H. B. Li, G. W. Ma, J. Zhao, Z. L. Zhou, and F. Dai (2012). “Suggested methods for determining the dynamic strength parameters and mode-I fracture toughness of rock materials”. In: *International Journal of Rock Mechanics and Mining Sciences* 49, pp. 105–112. DOI: 10.1016/j.ijrmms.2011.10.004.
- Zhu, Y., Y. Tang, H. Wang, and Q.-B. Zhang (2024). “Development of a dynamic cumulative damage model and its application to underground hydropower caverns under multiple blasting”. In: *International Journal of Rock Mechanics and Mining Sciences* 184, p. 105948. DOI: 10.1016/j.ijrmms.2024.105948.